4G 4G 2:49 47
vulms.vu.edu.pk/Quiz/Q 13

Time Left 81 sec(s)

MTH621 - Real Analysis I (Quiz # 3)

Quiz Start Time: 02:46 PM, 24 July 2023

Question # 5 of 5 (Start time: 02:49:05 PM, 24 July 2023)

Total Marks: 1

The function $g(x) = -x^3$ is decreasing on ------.

Select the correct option

Reload Math Equations

- $[0, \infty)$
- none of these
- $(-\infty, 0)$
- $(-\infty, \infty)$

Click to Save Answer & Move to Next Question

4G 4G 2:46
4G 48
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13

The function $h(x) = -x^3$ is

elect the correct option

- positive on $(-\infty, \infty)$.
- increasing on $(-\infty, \infty)$.
- is undefined on $(-\infty, \infty)$.
- decreasing on $(-\infty, \infty)$.

Click to Sa

4G 4G 2:44 4G 49
vulms.vu.edu.pk/Quiz/Q 13

Investigate the $\lim it\ of\ \lim_{x \to 0} 2x \sin \sqrt{x} = - - - -.$

Select the correct option

○	x
○	*none of these*
○	0
○	1

Click to Save Ans

Question # 2 of 5 (Start time: 02:43:39 PM, 24 July 2023) Total Marks: 1

Investigate the $\lim$ *it of* $\lim_{x \to 0} \frac{\sin x}{x} = - - - - -$.

Select the correct option

Reload Math Equations

- x
- *none of these*
- 0
- 1

Click to Save Answer & Move to Next Question

4G 4G 2:43
4G2 49
vulms.vu.edu.pk/Quiz/Q
13

Question # 1 of 5 (Start time: 02:42:52 PM, 24 July 2023)

Investigate the $\lim it\ of\ \lim_{x\to 0^-} \frac{1}{x} = ----.$

Select the correct option

○	∞
○	0
○	*none of these*
○	$-\infty$

Click to Save Answer &

MTH621 - Real Analysis I (Quiz # 3) Quiz Start Time: 02:20 PM, 24 July 2023

Question # 5 of 5 (Start time: 02:26:06 PM, 24 July 2023) Total Marks: 1

If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then f is

Select the correct option

Reload Math Equations

- continuous at c.
- having one sided limit.
- discontinuous at c.
- undefined at c.

Click to Save Answer & Move to Next Question

Question # 4 of 5 (Start time: 02:25:19 PM, 24 July 2023) Total Marks: 1

Investigate the $\lim it\ of\ \lim\limits_{x\to 0} x\sin\frac{1}{x} = ----.$

Select the correct option

Reload Math Equations

○	1
○	x
○	*none of these*
○	0

Click to Save Answer & Move to Next Question

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Quiz
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MTH621 - Real Analysis I (Quiz # 3)

Question # 4 of 5 (Start time: 02:22:54 PM, 24 July 2023)

The inverse of the function $f(x) = x^2$, is

Select the correct option

- $f^{-1}(y) = \sqrt{y}$.
- $f^{-1}(y) = \frac{1}{x^2}$
- $f^{-1}(y) = y^2$
- $f^{-1}(y) = \frac{1}{y^2}$

Question # 3 of 5 (Start time: 02:22:13 PM, 24 July 2023)

The function $g(x) = x^2$ is

Select the correct option

○	undefined on $[0, \infty)$.
○	decreasing on $[0, \infty)$.
○	having negative values on $[0, \infty)$.
○	increasing on $[0, \infty)$.

Question #1 of 5 (Start time: 02:21:00 PM, 24 July 2023) **Total Marks: 1**

The function $f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$ is

Select the correct option

Reload Math Equations

- is nondecreasing on $I = [0, 2]$.
- is undefined on $I = [0, 2]$.
- is nonincreasing on $I = [0, 2]$.
- is having negative values on $I = [0, 2]$.

Click to Save Answer & Move to Next Question

$x \to x_0$

$$\lim_{x \to 0-} \frac{1}{x} = -\infty,$$

$$\lim_{x \to 0+} \frac{1}{x} = \infty;$$

$$\lim_{x \to 0-} \frac{1}{x^2} = \lim_{x \to 0+} \frac{1}{x^2} = \infty$$

$$\lim_{x \to 0} \frac{1}{x^2} = \infty;$$

$$\lim x^2 = \lim_{x \to -\infty} x^2 = \infty;$$

Example: Prov... $\lim_{x \to 0+} \sqrt{x} = 0$.

Solution: See Lecture.

Example: Show that

$$\lim_{x \to 0+} \left(\frac{|x|}{x} + x \right) = 1, \qquad \lim_{x \to 0-} \left(\frac{|x|}{x} + x \right) = -1.$$

Solution: See Lecture

Theorem: A function f has a limit at x_0 if and only if it has left- and right-h... limits at x_0, and they are equal. More specifically,

$$f'(x_0) = \lim_{x \to x_0} \sum_{k=0} x^{n-k-1} x_0^k = n x_0^{n-1}.$$

ince this holds for every x_0, we drop the subscript and write

$$f'(x) = nx^{n-1} \quad \text{or} \quad \frac{d}{dx}(x^n) = nx^{n-1}.$$

ample: Find the derivative of the line $y = mx + c$.

lution:

Since R_g [illegible] theorem, we have proved with f and [illegible] implies that g is continuous on $[c, d]$.

Example: If

$$f(x) = x^2, \quad 0 \le x \le R.$$

Solution: The inverse of the given function is

$$f^{-1}(y) = g(y) = \sqrt{y}, \quad 0 \le y \le R^2.$$

Example: If

$$f(x) = 2x + 4, \quad 0 \le x \le 2,$$

Solution: The inverse of the given function is

$$f^{-1}(y) = g(y) = \frac{y-4}{2}, \quad 4 \le y \le 8.$$

(3.13)

In either case, f is on I.

If $\leq$ can be replaced by $<$ in (3.12), f is *increasing* on I. If $\geq$ can be replaced by $>$ in (3.13), f is *decreasing* on I. In either of these two cases, f is *strictly monotonic* on I.

Example: The function

$$f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$$

is nondecreasing on $I = [0, 2]$.

Example: The function $g(x) = x^2$ is increasing on $[0, \infty)$. The function $h(x) = -x^3$ is decreasing on $(-\infty, \infty)$.

Theorem: Suppose that f is monotonic on (a, b) and define

$\lim_{x \to 0+} \sqrt{x} = 0.$

Solution: See Lecture.

Example: Show that

$$\lim_{x \to 0+} \left(\frac{|x|}{x} + x \right) = 1, \qquad \lim_{x \to 0-} \left(\frac{|x|}{x} + x \right) = -1.$$

Solution: See Lecture

Theorem: A function f has a limit at x_0 if and only if it has left- and limits at x_0, and they are equal. More specifically,

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Example: The function

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 2, & 1 \le x \le 2, \end{cases}$$

is nondecreasing on $I = [0, 2]$.

Example: The function $g(x) = x^2$ is increasing on $[0, \infty)$. The function is decreasing on $(-\infty, \infty)$.

Theorem: Suppose that f is monotonic on (a, b) and define

$\alpha = \inf f(x)$ and $\beta = \sup f(x)$

Example: For the function

$$f(x) = x \sin \frac{1}{x}, \quad x \neq 0,$$

show that

$$\lim_{x \to 0} f(x) = 0$$

Solution: Even though f is not defined at $x_0 = 0$, because if

$$0 < |x| < \delta = \varepsilon,$$

then

$$|f(x) - 0| = \left| x \sin \frac{1}{x} \right| \leq |x| < \varepsilon.$$

On the other hand, the function

$$g(x) = \sin \frac{1}{x}, \quad x \neq 0,$$

has no limit as x approaches 0, since it assumes all values between -1 and 1 in every neighborhood of the origin.